PLANT PROTECTION DEPARTMENT

NATIONAL CENTER FOR FERTILIZER TESTING

PRODUCT DEMONSTRATION RESULTS REPORT

"Evaluation of the effect of Jipyeong products on rice plants"

Name of the organization performing the demonstration:	National Center For Fertilizer Testing
Name of the organization with demonstration product:	Booyang Material Co., Ltd
Location of the performance:	Nguyen Thanh Tai farmer household, Binh An commune, Thu Thua district, Long An province

Hanoi, 2024

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"Evaluation of the effect of Jipyeong products on rice plants"

1. GENERAL INFORMATION

1.1. Name of the organization with the demonstration product

- Name of the organization with demonstration product: Booyang Material Co., Ltd

1.2. Name and address of the organization of the performance

- Name of the organization performing the demonstration: National Center For Fertilizer Testing
- Address: No. 65 Sa Doi Street, Phu Do Ward, Nam Tu Liem District, Hanoi City

1.3. Name and origin of the demonstration product

- Product name: Jipyeong.
- Origin of the demonstration product: The product is manufactured by Booyang Material Co., Ltd.

1.4. Quality criteria according to the registration outline

Table 1.1 Quality criteria of demonstration products

No.	Fertilizer Name	Units of Calculation	Quality Criteria	
			Name of the indicator	Content
1	Jipyeong	%	*Radish juice*	97
		%	*Sulphur*	3

1.5. Types of products, modes of use, characteristics and main uses of demonstration products

Table 1.2 Characteristics of the product

No.	Fertilizer Name	Fertilizer Type	How to use	Characteristic	Main uses
1	Jipyeong	Foliar fertilizer	Foliar dressing	Liquid form	- Helps strengthen disease resistance and inhibit necessary external intrusion through stems and leaves for plants to help plants grow and develop well - Increase crop yields.

1.6. Grounds and requirements for performance

1.6.1. Grounds for conducting the demonstration

- Law on Crop Cultivation No. 31/2018/QH14 was approved by the 14th National Assembly, at the 6th session on 19/11/2018.

- Decree No. 84/2019/ND-CP dated November 14, 2019 of the Government regulating fertilizer management.

- QCVN 01-189:2019/BNNPTNT - National technical regulation on fertilizer quality (Promulgated under Circular No. 09/2019/TT-BNNPTNT dated August 27, 2019 of the Ministry of Agriculture and Rural Development)

- TCVN 12719:2019 testing fertilizers for annual crops.

- Based on the agreement signed between the two parties.

1.6.2. Purpose and requirements of the performance

- **Purpose of the demonstration**: To evaluate the disease resistance, inhibit the intrusion from the outside, avoid calcium deficiency to increase the crop yield of the product on rice.

- **Demonstration Requirements**: Evaluate the effect of Jipyeong products compared to recipes using local farming methods and recipes without Jipyeong products

2. DEMONSTRATION METHOD

2.1. Conditions for performance

2.1.1. Location of the performance

Nguyen Thanh Tai farmer household, Binh An commune, Thu Thua district, Long An province

2.1.2. Performance time

Table 2.1 Performance time

Seasonality	Show start date	Harvest date
Autumn-Winter Crop	06/08/2024	12/11/2024

2.1.3. Demonstration crops

- Demonstration crops: Rice plants
- Variety: IR 4625
- Area: 1.000m^2
- Seed amount used 100kg/ha

2.1.4. Farming regime

****) History of using the product at the demonstration site:***

At the demonstration venue (Thu Thua, Long An), the history of using the product for rice plants is as follows: DAP fertilizer 80 kg/ha + Urea fertilizer 160 kg/ha + NPK fertilizer 20-20-15 100 kg/ha + Potassium chloride 150 kg/ha (equivalent to N: 108 kg/ha + P_2O_5: 56,8 kg/ha + K_2O: 112,6 kg/ha).

2.1.5. Irrigation conditions and other technical measures.

+ Irrigation conditions: there is an active irrigation system, the main water source is surface water.

+ Other technical measures

- Soil preparation: The soil is plowed deeply and indiscriminately, then the axis is leveled and combined to make ditches for drainage.

- Sowing technique: Apply hand sowing technique. The amount of seeds according to the local process is 100 kg/ha. Soak the seeds for 24 hours. Pick up, wash thoroughly and compost the seeds. Sow when the seed sprouts 0,5-1cm.

- Spraying herbicides:

+ Weeding out grasses (pre-germination): 2-3 days after sowing.

+ Weeding out grasses and leaves (after germination): 7-10 days after sowing.

2.1.6. Weather and climatic conditions

Weather Conditions:

The weather situation during the performance in Long An province is shown in Table 2.2:

Table 2.2. Weather conditions during the performance in Long An province

No.	MONTH	TEMPERATURE	RAINFALL	HUMIDITY	HOURS OF SUNSHINE
1	08	28,4	60	79	256
2	09	27,8	261	83	163
3	10	27,9	551	82	208
4	11	27,8	186	79	230

Source: National Center for Hydrometeorological Forecasting

Climatic conditions:

The climate of Long An is located in the Mekong Delta, which belongs to the monsoon tropical climate, with characteristics. The average temperature is quite stable, ranging from 27,8°C to 28,4°C, August has the highest temperature (28,4°C), which is suitable for summer weather, and September and November have the lowest temperature (27,8°C), showing a trend of decreasing heat when the season changes. Rainfall fluctuates sharply through the months of August with the lowest rainfall (60 mm), dry weather, October with the highest rainfall (551 mm), showing that this is the heaviest rainy month, September (261 mm) and November (186 mm) have average rainfall, indicating the transition between the rainy and dry seasons. Relative humidity is high in all months, ranging from 79% to 83%, September has the highest humidity (83%), and is also one of the months with the most rain, August and November have the lowest humidity (79%), consistent with less rainfall. The number of sunshine hours tends to decrease as rainfall increases, August has the highest number of sunshine hours (256 hours), dry and sunny, September has the lowest number of sunshine hours (163 hours),

due to heavy rainfall, October and November have recovered sunshine hours of 208 hours and 230 hours, respectively, corresponding to a decrease in rainfall.

2.1.7. Comment on the situation of occurrence, development, pests and diseases, resistance of plants during the demonstration period

During the demonstration, it was shown that some of the following pests and diseases appeared on the rice plants:

- Pests: Leaf roller caterpillar (scientific name *Parnara guttata)*, brown leafhopper (scientific name *Nilaparvata lugens)*

- Disease: *Pyricularia oryzae*

2.2. Demonstration

2.2.1. Demonstration formula and control formula

+ Small scale (300 m^2)

CT 1 (Control 150 m^2): Traditional cultivation method

CT 2 (Jipyeong 150 m^2) : Booyang product + Traditional Farming Products

+ Large scale (700 m^2)

CT 1 (Control 350 m^2): Traditional cultivation method

CT 2 (Jipyeong 350 m^2) : Booyang product + Traditional Farming Products

2.2.2. Experimental design

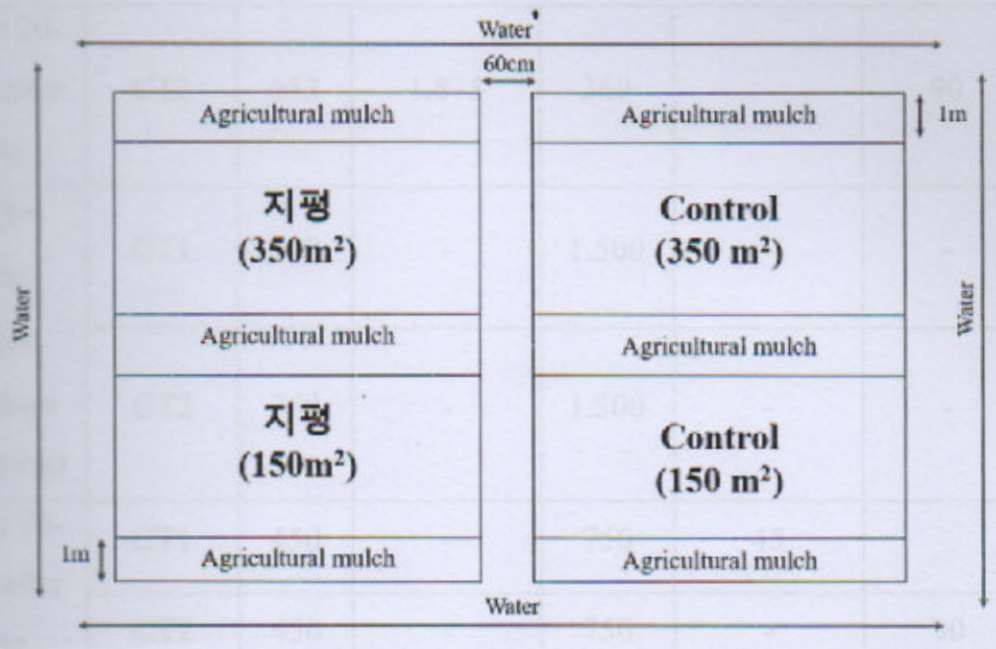
- The experiment is arranged on a plot of land in a non-repetitive sequential manner with 2 scales of each scale with 02 recipes, 01 recipe using the local farming method, and 01 recipe using the local farming method in combination with the demonstration product.

Area of each experimental cell:

+ Small scale: 150 m^2/cell. The total area of the experiment is 300 m^2 (excluding trenches and protection strips), the experiment is arranged on a land plot.

+ Large-scale: 350 m^2/cell. The total area of the experiment is 700 m^2 (excluding trenches and protection strips), the experiment is arranged on a plot of land.

- Experiment layout diagram:



2.2.3. Method of use

At the demonstration site, the spraying method of the experimental products is designed as follows:

Table 2.3. Fertilizer amount and fertilization period on rice plants of small-scale formulations at the demonstration site

Application, application period	Formula	Amount of fertilizer, g/150m²				
		Urea, g	Phosphate, g	KCl, g	Control: Water, liter	Jipyeong, ml
1st time: Seed soaking	CT1	-	-	-	15	-
	CT2	-	-	-	-	75
2nd time: Apply 2-3 days after sowing	CT1	450	-	-	-	-
	CT2	450	-	-	-	-
3rd time: 8-10 days after sowing	CT1	1.219	3.450	-	-	-
	CT2	1.219	3.450	-	-	-
	CT1	653	1.875	369	45	

4th time: 20-25 days after sowing	CT2	653	1.875	369	-	90
5th: Prior panicle initiation (40-45 days after sowing)	CT1	750	-	1.500	-	-
	CT2	750	-	1.500	-	-
6th time: 70-75 days after sowing	CT1	450	-	750	45	
	CT2	450	-	750	-	90

- **Fertilization technique:**

+ Fertilizer Jipyeong: The amount of fertilizer used each time in each recipe is mixed with 45 liters of water and sprayed for an area of 150m^2.

+ Fertilization time: Morning or cool afternoon

Table 2.4. Fertilizer amount and fertilization period on rice plants of large-scale formulations at the demonstration site

Application, application period	Formula	Amount of fertilizer, g/350m^2				
		Urea, g	Phosphate, g	KCl, g	Control: Water, liter	Jipyeong, ml
1st time: Seed soaking	CT1	-	-	-	30	-
	CT2	-	-	-	-	150
2nd time: Apply 2-3 days after sowing	CT1	1.050	-	-	-	-
	CT2	1.050	-	-	-	-
3rd time: 8-10 days after sowing	CT1	2.844	8.050	-	-	-
	CT2	2.844	8.050	-	-	-

4th time: 20-25 days after sowing	CT1	1.524	4.375	861	105	
	CT2	1.524	4.375	861	-	210
5th: Prior panicle initiation (40-45 days after sowing)	CT1	1.750	-	3.500	-	-
	CT2	1.750	-	3.500	-	-
6th time: 70-75 days after sowing	CT1	1.050	-	1.750	105	
	CT2	1.050	-	1.750	-	210

- **Fertilization technique:**

+ Fertilizer Jipyeong: The amount of fertilizer used each time in each recipe is mixed with 105 liters of water and sprayed for an area of 350m^2.

+ Fertilization time: Morning or cool afternoon.

2.2.4. Water and fertilizer regime

Table 2.5 Water regime for rice at the demonstration site

The day after sowing	Adjusting the water level in the field	Notes
0 - 7	The soil is just enough moisture or the water level is about 1cm higher than the field surface.	Drain the water before sowing and keep the field surface dry within 3 days after sowing, on the 4th day, put water into the field surface for 01 day; then drain to ensure sufficient moisture on the field surface.
7 - 10	Put water in the field 1 - 3cm to apply the 1st fertilizer, keep the water continuously until the 2nd fertilizer application (20-25 days after sowing).	Limit the germination of grasses.

The day after sowing	Adjusting the water level in the field	Notes
25 - 40	Pump water 3-5cm above the field surface when the water level in the pipe is 15cm below the ground.	The low water level limits the spread of fungal lymphadenopathy in the field, which is less contagious. The farther the water level below the ground (but not lower than 15 cm above the ground) will help the rice roots to penetrate deep into the soil, prevent falling and be easy to harvest.
40 - 45	Pump water to about 1 - 3cm to apply fertilizer (also known as fertilizer).	Pumping water before fertilizing limits the decomposition and evaporation of light fertilizers, especially nitrogen fertilizers.
65 - 75	Keep the water level in the field 3 - 5cm continuously for 10 days.	Helping rice plants mature, pollinate and fertilize easily, rice grains do not flatten and float.
75 days after sowing to harvest	Keep the water level from ground level to 15cm below the ground.	Squeeze the water 10 days before harvesting so that the field surface is dry, easy to apply with a harvester.

- The amount of fertilizer used during the whole testing process: DAP fertilizer 80 kg/ha + Urea fertilizer 160 kg/ha + NPK fertilizer 20-20-15 100 kg/ha + Potassium chloride 150 kg/ha (equivalent to N: 108 kg/ha + P_2O_5: 56,8 kg/ha + K_2O: 112,6 kg/ha).

- Plant protection:

Table 2.6 Types of pests and pesticides expected to be used in the experiment

Pests and diseases	Pesticides used
Leaf rollers, thrips	Incipio 200SC, Silsau 6.5EC
Brown leafhopper	Anproud 70WG, Lobby 25WP
Blast disease, smudge seeds	Amistar Top 325SC, Help 400SC

+ Pest control: Regularly inspect the field, monitor pests and diseases, and conduct prevention when necessary.

+ When pests and diseases appear, it is necessary to monitor and control according to the principle of "4 right" drugs, the right dosage – concentration, right time and right way to prevent pests and diseases.

2.2.5. ***Monitoring indicators and data collection methods***

- Monitoring indicators:

+ Growth and development of plants

+ Percentage of fine grains;

+ Volume of 1.000 seeds;

+ Crop yield;

+ Harvest crop yields.

- Data collection method:

+ Weather data: collected from agencies with hydrometeorological observation function

- **Growth and development targets:**

+ Time from sowing to prior panicle initiation (day): From the day of sowing to the day of the lamp heart (germ) 1-3mm.

+ Time from sowing to ripening (days): From the date of sowing to the day when 50% of the trees on the experimental cell are blooming.

+ Time to ripen (days): From the time of sowing to the time when there are 90% of the seeds on the ripe cotton.

- **Crop yield indicators:**

+ Percentage of firm seeds: select 10 random flowers in the experiment box to count the number of firm seeds and flat seeds for average calculation

+ Volume of 1.000 particles: 1.000 particles are randomly taken on the mass weighing laboratory cell

Crop yield (kg/ha): The crop yield in the demonstration is determined on the basis of harvesting the whole demonstration plot and then calculating the sum of the harvests converted to kg/ha.

- **Yield Harvest:**

+ Yield is calculated according to the formula:

$$BTns\ (\%) = \frac{(NS_{kn} - NS_{dc})}{NS_{dc}} \times 100$$

In which:

BT_{ns}: Yield harvest (unit: %);

NS_{kn}: Crop yield at the demonstration formula (unit: kg/ha);

NS_{dc}: Crop yield in the control formula (calculation unit: kg/ha).

3. PERFORMANCE RESULTS

3.1. Comment on weather conditions, climate, irrigation water regime for crops, growth and development of crops, pests and diseases

3.1.1 Weather and climatic conditions

During the performance, the average air temperature ranges from 27,8-28,4^{0}C (suitable for rice plants to grow and develop). Rainfall from 60-551 mm is relatively high for the growth and development of rice plants, but it is still necessary to manage water on rainy days without rain to ensure water for plants. Therefore, rice plants still grow and develop normally.

3.1.2 Pests and diseases

During the demonstration, it was shown that some of the following pests and diseases appeared on the rice plants:

- Pests: Leaf roller caterpillar (scientific name *Parnara guttata)*, brown leafhopper (scientific name *Nilaparvata lugens)*

- Disease: *Pyricularia oryzae*

Table 3.1 Pesticides used during the experiment.

Pests and diseases	Pesticides used
Leaf roll worm	Incipio 200SC
Brown leafhopper	Anproud 70WG
Blast disease	Amistar Top 325SC

However, pests and diseases all appear at a low level and are prevented in time, so it does not affect the growth and productivity of rice plants.

3.1.3 Growth and development period

Table 0.2. Growth and development time of rice plants through stages at the demonstration site

Unit: days

Formula	Prior panicle initiation	Flowering	Ripening
		Small Scale	
CT1 (Control)	42 days after sowing	69 days after sowing	94 days after sowing
CT2	42 days after sowing	69 days after sowing	94 days after sowing
		Large-scale	
CT1 (Control)	42 days after sowing	69 days after sowing	94 days after sowing
CT2	42 days after sowing	69 days after sowing	94 days after sowing

3.2. Percentage of firm grains at the laboratory

Table 3.3. Yield and yield bumper on rice plants in the experiment

Quota	Average number of firm seeds and number of flat seeds/cotton		Weight 1000 Grains
	Number of Firm Particles	Number of flat Grains	
		Small Scale	
CT1	63,2	11	42,87
CT2	65,6	10,6	44,76
		Large-scale	
CT1	62,7	11,4	42,23
CT2	65,1	10,1	44,52

Table 3. 3 shows :

- For small scale: Using the product Jipyeong (150m²), the number of solid particles is 65,6, the number of crushed particles is 10,6 and the weight of 1.000 grains

is 44,76.Under the control condition (150m²), the number of solid particles decreases to 63,2 the number of crushed particles increases slightly to 11 and the weight of 1.000 grains also decreases to 42,87. Thus using the "Jipyeong" product, the indicators are better than Control: the number of strong particles is higher, the number of particles is smaller, and the weight of 1.000 grains is also higher.

- For large scale: Using Jipyeong (350m²), the number of solid particles is 65.1, the number of crushed particles is 10,1 and the weight of 1000 grains is 44,52. Under the control condition (350m²), the number of solid particles decreases to 62,7 the number of crushed particles increases to 11,4 and the weight of 1.000 grains decreases to 42,23. Thus Similar to the 150m² condition, using the product Jipyeong, the indicators are better Control: the number of strong grains is more, the number of small grains is less, and the weight of 1.000 grains is also higher.

- In summary, the formula using the product Jipyeong (area of 150m² and 350m²) showed superior results to Control in all indicators (higher number of firm particles, lower number of small particles and greater weight of 1.000 particles). The difference between productJipyeong and control is stable in both area ranges, indicating the effectiveness of productJipyeong use conditions in improving grain yield.

3.3. Yield and bumper yield of rice plants

Table 3.4. Yield and yield bumper on rice plants in the experiment

Quota	Total productivity (kg)	Yield overyield compared to control (%)
	Small Scale	
CT1	79	-
CT2	87	10,13
	Large-scale	
CT1	185	-
CT2	204	10,27

Table 3.4 shows the Small-Scale Experiment: at the Control formula (150 m²) the yield reaches 79 kg. the formula uses the product for a yield of 87 kg (yield bumper

compared to the control formula is 10,13%). Large-scale experiments: the Control formula (350 m^2) yielded 185 kg, the formula used the product yielded 204 kg (yield yield compared to the control formula was 10,27%). We found that the yield of the formula using the product was >10% higher than the control formula at both experimental scales, indicating the effect of increasing the yield of the product Jipyeong in the demonstration experiment.

3.4. Sulfur (S) test results on rice plants

Table 3.5 Sulfur (S) test results on rice

Quota	S (mg/kg)
Small scale	
CT1	100,83
CT2	113,58
Large scale	
CT1	106,55
CT2	109,34

Table 3.5 shows:

- CT2 yielded higher sulphur content than CT1 at both scales, demonstrating that CT2 may be a more efficient option in terms of sulphur accumulation.

- However, the difference between small and large scale is not too large, indicating the stability of the test formulas.

4. CONCLUSIONS AND RECOMMENDATIONS

4.1. Conclusion

- Through the process of small-scale and large-scale demonstrations, the product Jipyeong (spraying level 0,5 liters/ha/time) in Binh An Commune, Thu Thua District, Long An Province has a higher yield and yield (of both scales) than the formula without using the product, specifically as follows:

- Small-scale demonstration with actual yield of 87 kg (yield yield compared to control is 10,13%).

- Large-scale demonstration with actual yield of 204 kg (yield yield compared to control is 10,27%).

- Thus, the results of the demonstration of narrow and wide area on rice plants of Jippyeong product (spraying level 0,5 liters/ha/time) had the effect of increasing the actual yield by 10% higher than the control.

- Based on the demonstration results, it can be seen that the effectiveness of using Jippyeong products combined with local farming methods can be seen. Yields higher yields than recipes that only use local farming methods. Meanwhile, the cost of use and labor of the combination of jiping and local farming methods is still high compared to the local farming method.

4.2. Recommendations

- The test results show that the formula using high yield of 10,13% fertilizer meets the TCVN, request the official testing company according to Vietnamese regulations for detailed evaluation, if the test results still meet the TCVN, then request the issuance of a circulation decision in Vietnam.

- People's petitions for products that are used many times increase labor, dosage and use is still a lot.

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TRUNG TÂM KHẢO NGHIỆM VÀ KIỂM NGHIỆM PHÂN BÓN QUỐC GIA

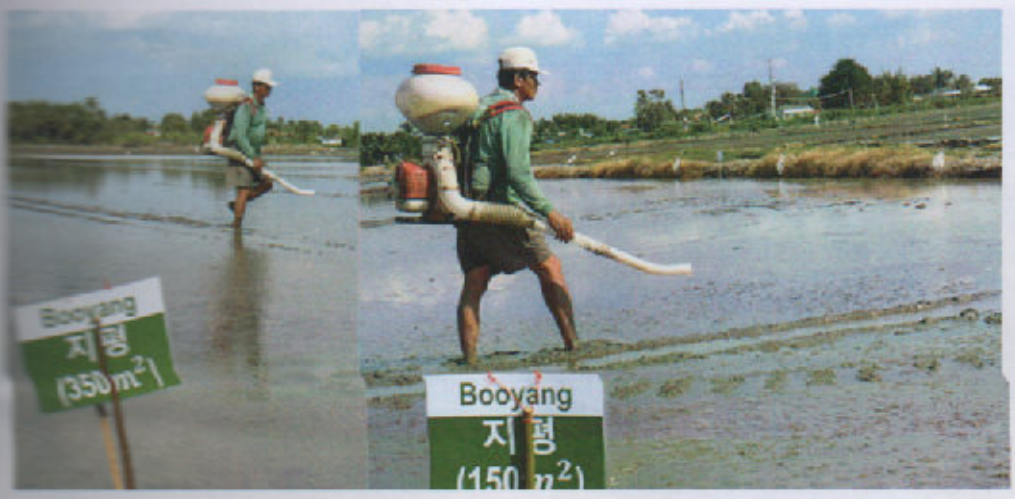
GIÁM ĐỐC

NGUYỄN THỊ KIM DUNG

PROCESS IMAGE

Experimental layout image

Image of fractional processing 1



Seeding image

Image of 2nd Batch Processing

Image of 3rd Batch Processing

Harvest Images